



ISLAMIC UNIVERSITY OF TECHNOLOGY

EEE 4604

Measurement and Instrumentation LAB

Name of the Project : HOME AUTOMATION AND MONITORING SYSTEM

Group Name : TENACIOUS

Team Member	Student ID
Ar Rafi Ishraq	210021330
Ishrak Anan	210021212
Taufeeq Hassan Omio	210021214
Ashfiq Ul Rahman	210021342
Mir Mushfiq Rahman Ramim	210021253
Asiful Hoque Farabi	210021251

Introduction :

The rapid growth of embedded systems and smart sensors has revolutionized how environmental parameters are monitored and controlled in both industrial and domestic settings. This project, titled "Smart Environmental Monitoring and Automation System," is designed as a comprehensive implementation of real-time sensing, decision-making, and actuation using microcontroller-based instrumentation.

The objective of this project is to develop an automated system that senses multiple environmental parameters—such as light intensity, temperature, smoke concentration, water level, and object proximity—and responds appropriately through actuators like a fan, pump, and buzzer. The system employs a combination of sensors including LDR, DHT11, MQ-2, ultrasonic sensor (HC-SR04), and an IR proximity sensor, all interfaced with an Arduino Uno microcontroller. The outputs are managed using a relay module, buzzer, digital display, and user input devices such as a push button and potentiometer.

In addition to real-time monitoring, the system features safety and convenience functions. For example, the fan automatically turns on if the temperature exceeds a threshold, and a buzzer is triggered in case of smoke detection—along with a manual reset function. The pump is controlled based on object detection and water level feedback. A 16x2 LCD with I²C interface displays relevant data such as temperature and water level to the user.

This project demonstrates key concepts from the Measurements and Instrumentation course, including sensor calibration, signal interfacing, power management, digital control, and human-machine interaction. It provides a hands-on understanding of how sensors, microcontrollers, and output devices can be integrated to form intelligent automated systems.

Components & Properties :

- Microcontroller
 - Arduino Uno R3

- Sensors
 - LDR (Light Dependent Resistor)
 - DHT11 Temperature Sensor
 - MQ-2 Smoke Sensor
 - HC-SR04 Ultrasonic Sensor
 - IR Proximity Sensor

- Actuators and Outputs
 - DC Fan Motor (12V)
 - Mini Pump Motor (3V)
 - 2-Channel Relay Module
 - Active Buzzer
 - LCD 16×2 with I²C

- User Input Components
 - Push Button (SPST)
 - Potentiometer (10kΩ)

- Power Supply
 - 12V LiPo Battery
 - 3V AA Battery Pack (2 × 1.5V)

- Passive Components
 - Resistors (10kΩ, 100kΩ)
 - Diodes (1N4007, 1N5819)
 - Wires and Breadboard

Working Principle:

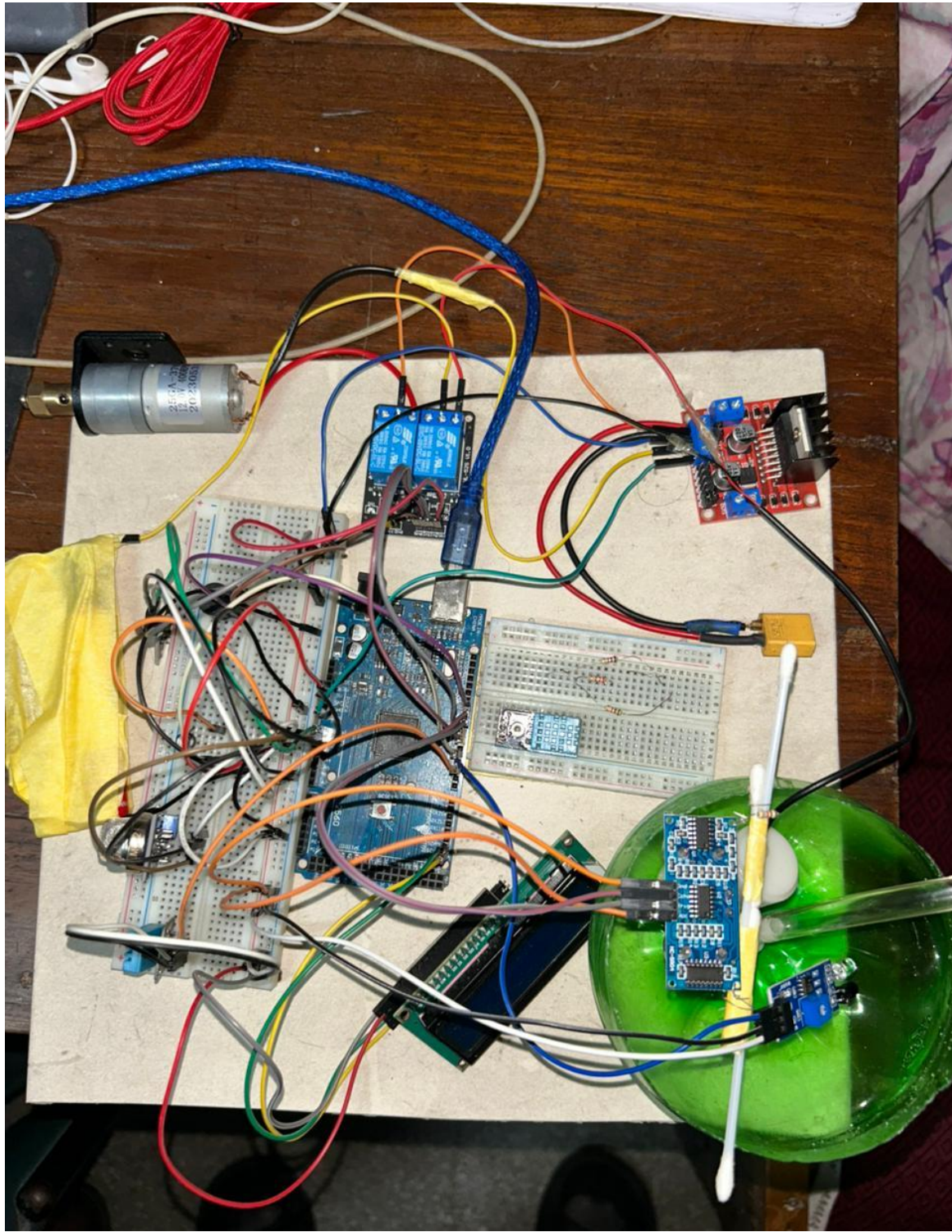
The Home Automation and Monitoring System operates by using various sensors and actuators to automate tasks and monitor environmental parameters. The system is activated when the LDR sensor detects low light conditions, such as in the evening, triggering the system to begin its monitoring functions. The DHT11 temperature sensor continuously measures the room temperature, and if it rises above a preset threshold (e.g., 30°C), the fan motor is turned on via a relay to provide cooling.

The system also incorporates smoke detection using the MQ-2 sensor. If smoke or gas is detected, an alarm buzzer is triggered to alert the user. The buzzer can be manually reset by pressing a push button. In addition, an ultrasonic sensor (HC-SR04) measures the water level in a tank, and the result is displayed on an LCD screen in real-time.

The IR proximity sensor detects the presence of an object or hand, and when detected, the 3V pump motor is activated to dispense water. Once the object is removed, the pump is automatically turned off. Users can adjust various system settings through a potentiometer, which can be used for threshold calibration or other system adjustments. The LCD display shows real-time data, while the push button can be used for manual interventions, such as resetting the buzzer.

The system is powered by a 12V LiPo battery for the Arduino and actuators, and a separate 2 × 1.5V AA battery pack is used to power the 3V pump. This setup ensures that the system can function autonomously based on environmental conditions such as light, temperature, smoke, and proximity, providing automated control over various household devices and monitoring tasks.

Project Picture :



Project Videos :

Click here : [Project Video](#)

Code :

```
1 // Include necessary libraries
2 #include <Wire.h>
3 #include <LiquidCrystal_I2C.h>
4 #include <DHT.h>
5
6 //--- PIN DEFINITIONS ---//
7 // Sensors
8 #define LDR_PIN A0
9 #define DHT_PIN 2
10 #define SMOKE_SENSOR_PIN A1
11 #define IR_SENSOR_PIN 3
12 #define TRIG_PIN 8
13 #define ECHO_PIN 9
14 // Actuators & Inputs
15 #define BUZZER_PIN 10
16 #define RESET_BUTTON_PIN 4
17 #define FAN_MOTOR_RELAY_PIN 6 // Relay IN1 for 12V Fan Motor
18 #define PUMP_MOTOR_RELAY_PIN 7 // Relay IN2 for 3V Pump Motor
19
20 //--- SENSOR & MODULE SETUP ---//
21 // LCD setup (address 0x27 is common, but may vary. Use an I2C scanner if it doesn't work)
22 LiquidCrystal_I2C lcd(0x27, 16, 2);
23
24 // DHT Sensor setup
25 #define DHT_TYPE DHT11
26 DHT dht(DHT_PIN, DHT_TYPE);
27
28 //--- PROJECT THRESHOLDS & SETTINGS (ADJUST THESE VALUES) ---//
29 const int LDR_ACTIVATION_THRESHOLD = 750; // Value below which the system activates (in darkness)
30 const float TEMPERATURE_THRESHOLD = 28.0; // Temperature in Celsius to turn on the fan
31 const int SMOKE_THRESHOLD = 400; // Analog value from MQ-2 to trigger alarm
32 const float TANK_HEIGHT_CM = 20.0; // Total height of your water tank in cm
33
34 //--- GLOBAL VARIABLES ---//
35 bool isSystemActive = false;
36 bool isSmokeAlarmActive = false;
```



```
37
38 void setup() {
39     // Start serial communication for debugging
40     Serial.begin(9600);
41
42     // Initialize LCD and DHT Sensor
43     lcd.init();
44     lcd.backlight();
45     dht.begin();
46
47     // Set pin modes for all components
48     pinMode(LDR_PIN, INPUT);
49     pinMode(SMOKE_SENSOR_PIN, INPUT);
50     pinMode(IR_SENSOR_PIN, INPUT);
51     pinMode(RESET_BUTTON_PIN, INPUT_PULLUP); // Use internal pull-up resistor
52
53     pinMode(TRIG_PIN, OUTPUT);
54     pinMode(ECHO_PIN, INPUT);
55
56     pinMode(FAN_MOTOR_RELAY_PIN, OUTPUT);
57     pinMode(PUMP_MOTOR_RELAY_PIN, OUTPUT);
58     pinMode(BUZZER_PIN, OUTPUT);
59
60     // Ensure all actuators are off at the start
61     digitalWrite(FAN_MOTOR_RELAY_PIN, HIGH); // Relays are often active-LOW
62     digitalWrite(PUMP_MOTOR_RELAY_PIN, HIGH);
63     digitalWrite(BUZZER_PIN, LOW);
64
65     // Initial message on LCD
66     lcd.setCursor(0, 0);
67     lcd.print("Home Automation");
68     lcd.setCursor(0, 1);
69     lcd.print("System Standby");
```

```
70     delay(2000);
71     lcd.clear();
72 }
73
74 void loop() {
75     // 1. Check for system activation via LDR
76     int ldrValue = analogRead(LDR_PIN);
77     Serial.print("LDR Value: ");
78     Serial.println(ldrValue);
79
80     if (ldrValue > LDR_ACTIVATION_THRESHOLD) {
81         if (!isSystemActive) {
82             isSystemActive = true;
83             Serial.println("System Activated");
84             lcd.clear();
85         }
86     } else {
87         if (isSystemActive) {
88             isSystemActive = false;
89             deactivateAllSystems();
90             Serial.println("System Deactivated");
91         }
92     }
93
94     // 2. Run main functions only if the system is active
95     if (isSystemActive) {
96         handleTemperature();
97         handleSmoke();
98         handleWaterDispenser();
99         displayWaterLevel();
100     } else {
101         // Display standby message if system is off
102         lcd.setCursor(0, 0);
```

```
103     lcd.print("System Standby ");
104     lcd.setCursor(0, 1);
105     lcd.print("Place obstacle ");
106 }
107
108     delay(500); // Main loop delay
109 }
110
111 // --- FUNCTION DEFINITIONS --- //
112
113 void handleTemperature() {
114     float temp = dht.readTemperature();
115
116     // Check if read failed
117     if (isnan(temp)) {
118         Serial.println("Failed to read from DHT sensor!");
119         return;
120     }
121
122     Serial.print("Temperature: ");
123     Serial.print(temp);
124     Serial.println(" C");
125
126     if (temp > TEMPERATURE_THRESHOLD) {
127         digitalWrite(FAN_MOTOR_RELAY_PIN, LOW); // Turn fan ON (active-LOW)
128         Serial.println("Fan ON");
129     } else {
130         digitalWrite(FAN_MOTOR_RELAY_PIN, HIGH); // Turn fan OFF
131         Serial.println("Fan OFF");
132     }
133 }
134
```

```

134
135 void handleSmoke() {
136     int smokeValue = analogRead(SMOKE_SENSOR_PIN);
137     Serial.print("Smoke Value: ");
138     Serial.println(smokeValue);
139
140     // Trigger alarm if smoke detected and alarm isn't already active
141     if (smokeValue > SMOKE_THRESHOLD && !isSmokeAlarmActive) {
142         isSmokeAlarmActive = true;
143         Serial.println("SMOKE DETECTED!");
144     }
145
146     // Sound the buzzer if the alarm is active
147     if (isSmokeAlarmActive) {
148         digitalWrite(BUZZER_PIN, HIGH);
149     } else {
150         digitalWrite(BUZZER_PIN, LOW);
151     }
152
153     // Check for reset button press to silence the alarm
154     if (digitalRead(RESET_BUTTON_PIN) == LOW && isSmokeAlarmActive) {
155         isSmokeAlarmActive = false;
156         digitalWrite(BUZZER_PIN, LOW);
157         Serial.println("Buzzer Reset");
158         delay(200); // Debounce delay
159     }
160 }
161
162 void handleWaterDispenser() {
163     // Corrected logic: IR sensor LOW = object detected -> Pump ON
164     if (digitalRead(IR_SENSOR_PIN) == LOW) {
165         digitalWrite(PUMP_MOTOR_RELAY_PIN, LOW); // Pump ON (active-LOW relay)
166         Serial.println("Object detected -> Pump ON");
167     } else {
168         digitalWrite(PUMP_MOTOR_RELAY_PIN, HIGH); // Pump OFF
169         Serial.println("No object -> Pump OFF");

```

```
170     }  
171 }  
172  
173 void displayWaterLevel() {  
174     // Trigger the ultrasonic sensor  
175     digitalWrite(TRIG_PIN, LOW);  
176     delayMicroseconds(2);  
177     digitalWrite(TRIG_PIN, HIGH);  
178     delayMicroseconds(10);  
179     digitalWrite(TRIG_PIN, LOW);  
180  
181     // Read the echo pulse  
182     long duration = pulseIn(ECHO_PIN, HIGH, 20000UL); // 20ms timeout  
183  
184     // Calculate distance in cm  
185     float distance = duration * 0.034 / 2;  
186  
187     // Apply your formula: (7 - distance)  
188     float adjustedValue = 7.0 - distance;  
189  
190     // Print to Serial Monitor  
191     Serial.print("Raw Distance: ");  
192     Serial.print(distance);  
193     Serial.print(" cm | Adjusted: ");  
194     Serial.print(adjustedValue);  
195     Serial.println(" cm");  
196  
197     // Display on LCD  
198     lcd.setCursor(0, 0);  
199     lcd.print("Adjusted Value "); // Clear line  
200     lcd.setCursor(0, 1);  
201     lcd.print("Dist: " + String(adjustedValue, 1) + " cm ");  
202 }  
203  
204  
205
```

```
205  
206 void deactivateAllSystems() {  
207     digitalWrite(FAN_MOTOR_RELAY_PIN, HIGH); // Fan OFF  
208     digitalWrite(PUMP_MOTOR_RELAY_PIN, HIGH); // Pump OFF  
209     digitalWrite(BUZZER_PIN, LOW); // Buzzer OFF  
210     isSmokeAlarmActive = false;  
211     lcd.clear();  
212 }  
213
```

Features :

Main Features of the Project:

- Light-Based System Activation (LDR)
- The system activates automatically in low light (e.g., evening or darkness).
- Uses an LDR sensor to detect ambient light.
- Temperature Monitoring & Cooling Control (DHT11 + Fan)
- Monitors room temperature in real-time.
- Automatically turns on a fan via a relay if temperature exceeds a set threshold (e.g., 30°C).
- Smoke Detection & Buzzer Alert (MQ-2 + Buzzer + Push Button)
- Detects smoke using the MQ-2 gas sensor.
- Triggers an audible buzzer alarm when smoke is detected.
- Buzzer can be manually reset using a push button.
- Smart Water Level Monitoring (Ultrasonic Sensor)
- Continuously monitors water level in a tank or container using an HC-SR04 sensor. Displays water level (in cm) on an LCD display.
- Contactless Water Dispensing (IR Sensor + Pump Motor)
- Detects the presence of a hand or object using an IR proximity sensor.
- Automatically turns on a small 3V pump to dispense water (e.g., for handwashing or filling a bottle). Stops pump when object is removed.
- LCD Display (I²C LCD 16×2)
- Real-time feedback: temperature, water level, system status.
- Easy to read, user-friendly interface.
- User Interface Inputs
- Potentiometer: can be used to tune thresholds or settings.
- Push button: for manually silencing the alarm (buzzer).
- Efficient Power Management
- Arduino powered from a 12V LiPo battery.
- Fan powered directly from 12V source.
- Pump powered by a separate 3V battery pack (2 × 1.5V AA).
- Relays isolate high-power loads from the Arduino.
- Modular & Expandable
- Code and hardware are modular — easy to add more sensors (e.g., humidity, flame, motion).
- Extra pins (D11–D13, A2–A3) still available.

- Cost-Effective & Educational
- Built entirely from low-cost components.

Demonstrates real-world use of sensors, actuators, and embedded systems principles.

Conclusion :

The Home Automation and Monitoring System effectively integrates a variety of sensors and actuators to automate and monitor critical environmental parameters within a home or workspace. The system provides functionalities such as light-based activation, temperature control, smoke detection, water level monitoring, and automatic water dispensing, offering an efficient, low-cost solution for home automation. By using sensors like the LDR, DHT11, MQ-2, and HC-SR04, and actuators like relays, motors, and buzzers, the system improves convenience and safety while providing real-time feedback via an LCD display. Additionally, the push button and potentiometer offer manual control for users, enhancing user interaction. The system is powered by a 12V LiPo battery and a separate 3V AA battery pack, ensuring reliable operation for both low-power and high-power components. This project not only highlights the practical use of embedded systems but also showcases the potential for scalability and future expansions with additional sensors or functionalities. Overall, it provides an excellent introduction to home automation, sensor integration, and actuator control, with applications in real-world smart homes or automation systems.